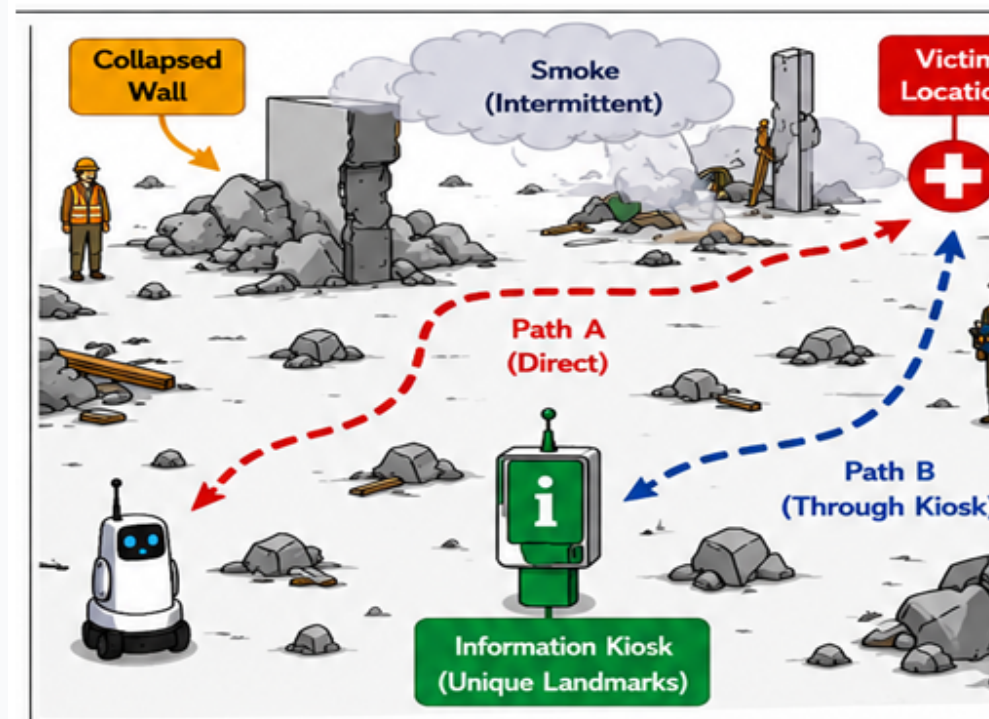


Question

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Use Case: A rescue robot is deployed inside a partially collapsed shopping complex after an earthquake. The robot possesses a pre-disaster map and uses a 2D LiDAR for localization and navigation. The environment has changed significantly due to collapsed walls, debris piles, smoke, and temporary rescue structures. Many landmarks stored in the original map are missing or damaged. However, an information kiosk containing a distinctive structure of landmarks remains intact and can potentially aid localization. As the robot navigates toward trapped victims to deliver aid, it continuously acquires sensor measurements and updates local occupancy estimates of the altered environment.



For simplicity, consider a **1-dimensional corridor model**.

The pre-disaster map contained two landmarks:

Information Kiosk : $L_1 = 10m$

Elevator Lobby nearby : $L_2 = 20m$

Robot initial pose:

$x_0 = 5m$

Assume below and use only this for mapping :

- Motion noise $Q = 1m$
- Sensor noise $R = 2m$
- Kalman Gain $K = 1$
- “**Fixed**” covariance inverse $S^{-1} = 0.5$

a) Apply One Iteration of FastSLAM with step by step illustration with two (assumptions) hypotheses before motion model is applied as Particle P1 : 6m Vs Particle P2 : 4m.

Post applying a control of $u=+3m$, LiDAR scan yielded two measurement points relative to the global frame: $Z1 = 1m$, $Z2 = 12m$. Use only the below simplified model to show one iteration of algorithm. Clearly show the updated landmark positions w.r.t each of particles P1 & P2 respectively and particle weights before resampling.

Simplified Model:

Landmark $_{NewPose} = \text{Landmark}_{OldPose} + \text{Kalman (Error)}$

Weight_{particle} = $e^{-\left(\sum_{\text{For every landmark}} (\text{Error} * \text{Error} * \text{CovarianceInverse})\right)}$

b) Justify your answer in no more than four well-articulated sentences valid reasoning.

- i. Modify the given rescue-robot scenario with suitable and realistic assumptions so that GraphSLAM becomes more appropriate than FastSLAM. Clearly explain why GraphSLAM would provide a better solution than FastSLAM
- ii. Modify the given rescue-robot scenario with suitable and realistic assumptions so that Occupancy Grid Mapping becomes more appropriate than SLAM. Clearly explain, why robot localization is no longer a primary problem, and why occupancy grid mapping alone is sufficient w.r.t your modified use case.

Note : *Vague answers will be penalized in the marking. Answers/Justifications must be articulated w.r.t the given use case only.*

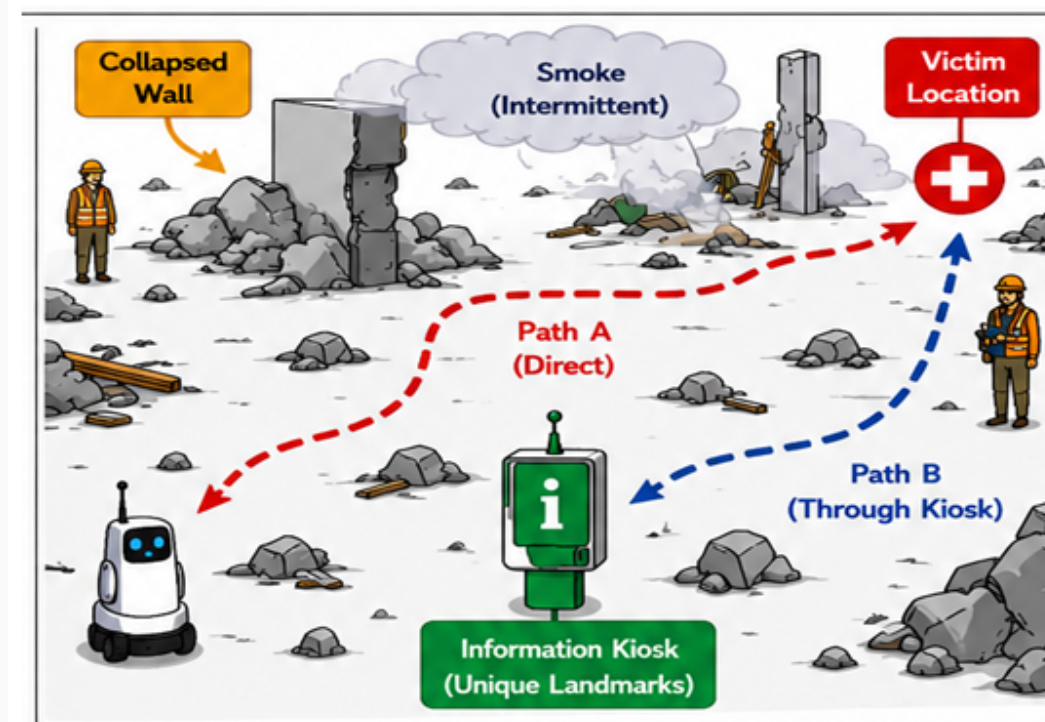
[6+2+2 = 10 Marks]

Question

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Note : *Vague answers will be penalized in the marking. Answers/Justifications must be articulated w.r.t the given use case only. For theoretical design questions, your answer must be in no more than five well-formed sentences.*

a) The rescue robot must travel from a rescue command post to the location of trapped victims within the damaged shopping complex. Several alternative navigation routes exist through partially blocked corridors. Assume that a Ant Colony Optimization (ACO) algorithm is used to identify promising routes. Five ants (with precomputed total cost) as depicted below are employed. Each ant traverses exactly one intermediate waypoint before reaching the goal. Apply only one iteration of ACO with step by step complete illustration using:

Initial pheromone on every route : $\tau=1$

Evaporation rate: $\rho=0.2$

Ant1 : Cost (Start à A à Goal) = 10

Ant2 : Cost (Start à B à Goal) = 8

Ant3 : Cost (Start à C à Goal) = 5

Ant4 : Cost (Start à D à Goal) = 12

Ant5 : Cost (Start à E à Goal) = 15

After one iteration Based on the updated pheromone values obtained, determine the first waypoint that would be selected by the next ant. Justify your answer.

b) Suppose the rescue-route planning problem is solved using a Genetic Algorithm (GA) instead of ACO. Design a chromosome representation, a gene representation, a suitable fitness function. Your fitness function must be different from the simple path-cost based fitness used in Part (a). Clearly justify your choices.

c) Create a modified version of the rescue-robot scenario where Deep Reinforcement Learning (DRL) becomes necessary. Justify and clearly state a multi-objective reward function, the purpose of each reward component. Justify your design choices.

[4+3+3 = 10 Marks]

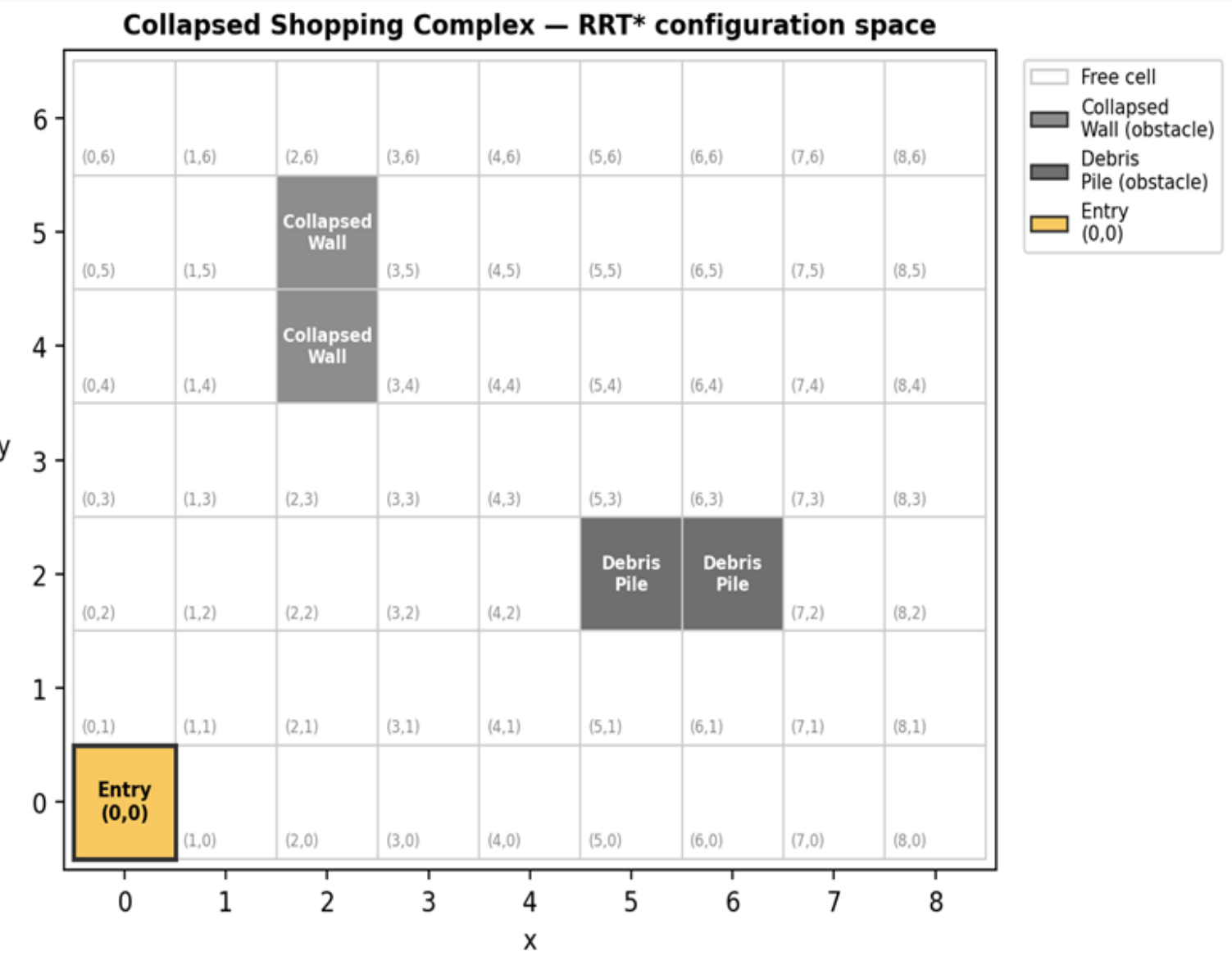
Question

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Use case: A rescue robot enters the partially collapsed shopping complex shown below (a collapsed wall and a debris pile block direct passage) to search for a trapped victim. It plans with Incremental RRT*.

Note: Answer parts (a)–(c) for the scenario below. Show every nearest-node search, steer computation, radius, candidate set, and cost. Vague answers will be penalized in the marking. Answers/Justifications must be articulated w.r.t the given use case only. For theoretical design questions, your answer must be in no more than five well-formed sentences



Configuration: Start (entry) = (0, 0), step size $\eta = 4$, $\gamma = 7$, dimension $d = 2$. Obstacle cells: Collapsed Wall = (2,4), (2,5); Debris Pile = (5,2), (6,2). Use Manhattan distance throughout.

Iteration	1	2	3	4
Sample x_{rand}	(5, 6)	(3, 2)	(5, 2)	(2, 2)

Collision rule: a sample is rejected if it coincides with a Collapsed-Wall or Debris-Pile cell.

Steer function, radius, and parent-and-rewire rule: Steer x-gap first then y-gap up to η ;

$r(n) = \min(\gamma \cdot (\ln n/n)^{(1/d)}, \eta)$

Note : Connect to whichever of $x_{nearest}$ or any node within $r(n)$ gives the lowest cost-to-come; then rewire any other in-radius node that strictly improves through x_{new}).

- a) Execute the learning phase of RRT* for all four iterations. For each: identify $x_{nearest}$, compute x_{new} , compute $r(n)$, list the candidate parent set, state the chosen parent and cost, and check for any rewiring (state explicitly if none occurs).
- b) Is RRT* probabilistically complete? State precisely what this guarantees and doesn't about eventually reaching the victim. If your resultant tree from part a) never got there, state one practical consequence of stopping this early, given the mission's urgency.
- c) Assume a modified scenario now, where several robots must search different debris zones for multiple victims simultaneously, instead of one tree toward one victim. Compare RRT* with Ant Colony Optimization (ACO) for this multi-robot mission: which is better suited, and why? Justify referring to how each scales to simultaneous searches and adapts as the debris field shifts.

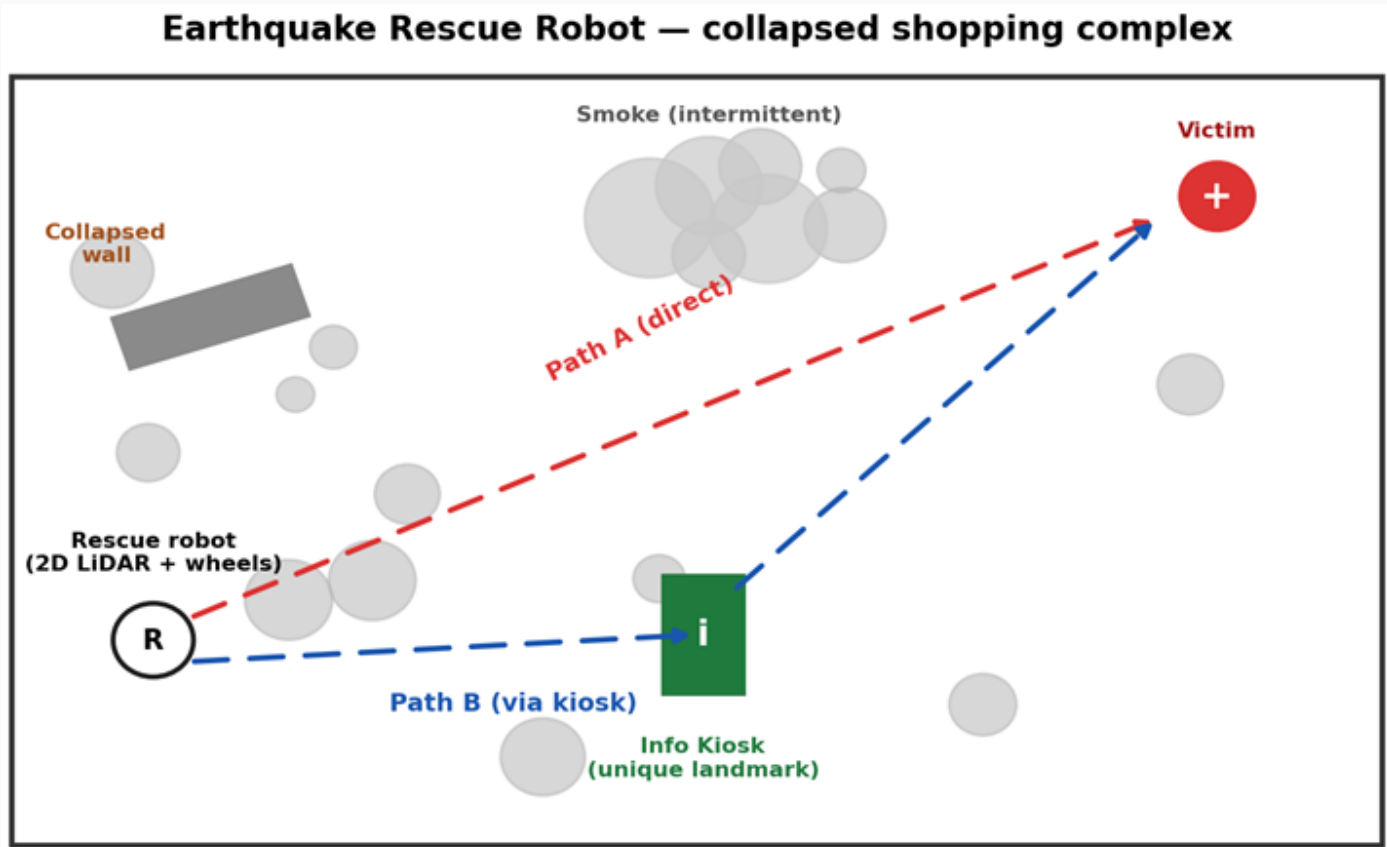
[5+2+2 = 9 Marks]

Question

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Use case: A rescue robot is deployed inside a partially collapsed shopping complex after an earthquake. It carries a pre-disaster map and navigates using a 2D LiDAR over wheeled locomotion across rubble. The environment has changed significantly collapsed walls, debris piles, and intermittent smoke that partially blocks scans but an information kiosk with a distinctive, intact landmark remains usable for localization. The robot must reach a victim, and may unknowingly be picked up and moved by a rescue worker while still powered on.



For the given use case, decide which technique in each pair below is best preferred over the other. Justify each answer with precise reasoning in no more than three well-formed sentences, tailored to this use case only.

Note : *Even though both options in a pair may be workable, only one is the best fit for this scenario. Vague, generic, or reference-material-copied answers will be penalized.*

- a) Velocity-based motion model vs Odometry-based motion model
- b) Beam-based (ray-cast) measurement model vs Endpoint (likelihood-field) measurement model
- c) Standard Monte Carlo Localization (MCL) vs Augmented MCL
- d) Bayesian (log-odds) occupancy-grid update vs Deterministic (binary) occupancy-grid update

[1.5x4 = 6 Marks]

